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Enhancing Cementing Operations Efficiency and Reliability by Utilizing Fiberglass-Based Non-Metallic Pipes for Plugging and Abandonment of Long Open Hole Intervals and Severe Loss Zones

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Abstract

This paper introduces an innovative technique for placing long cement plugs to abandon multiple zones in open hole sections, using fiberglass-based non-metallic tail pipes. Traditionally, the length of cement plugs was limited by the risk of metallic pipes getting stuck, necessitating multiple plugs for long intervals to isolate and address losses during drilling. By adopting this cost-effective solution of replacing conventional tail pipes with non-metallic pipes, the length of the cement plugs can be significantly increased, thereby avoiding the operational risks previously associated with metallic pipes. This new approach provides a reliable and efficient solution for plug and abandonment operations, as well as curing losses using large volume cement plugs. Utilizing metallic pipe and conventional placement techniques limits the volume of the cement plugs due to constraints on plug length, often resulting in multiple, repeated attempts before losses are cured or minimized.

The optimization of operational time by combining a P&A plug and kick-off plug (KOP) in a single attempt with a long cement plug placed across a highly deviated open hole using a fiberglass-based non-metallic tailpipe is discussed in this paper. Both pilot mother-bore abandonment and sidetracking into new laterals were achieved using conventional plug placement techniques, significantly improving well delivery objectives. Secondly, severe losses were cured across the reservoir section during drilling after pumping a large volume of cement plug. Finally, using non-metallic pipes to spot LCPs and/or P&A plugs showed a superior performance in a short period of time, which in turn will contribute towards reducing the associated non-productive time and further unwanted consequences of performing successive cement plugs. This paper explores the innovative use of sacrificial, fiberglass pipes to be able to pump long & large volumes of cement plugs. This technique has demonstrated significant advantages in preventing cement plug failures or the need for multiple plugs, ultimately optimizing rig time and reducing operational costs associated with drilling wells.

Introduction

One of the major operational risks associated with the placement of long cement plugs is the need to pull out the metallic tailpipe successfully after placement. Leaving these pipes inside the cement plug can lead to significant operational complications and ultimately substantial non-productive time (NPT) or, in some cases, loss of a well. The complications of a stuck pipe have historically forced operators to limit the length of cement plugs, often compromising zonal isolation effectiveness and requiring multiple, costly attempts to achieve full wellbore abandonment with cement plugs. These repeated operations not only consume rig time but also expose the well to increased risks, such as losses and incomplete zonal isolation.

To address these long-standing challenges, a novel technical approach has been introduced: the use of sacrificial fiberglass-based non-metallic tailpipes during long cement plug placement for different objectives. This technique allows the long cement plug to be in place without concern for future well integrity, eliminating the primary risk associated with metallic pipes. Fiberglass pipes offer excellent mechanical strength while being lightweight, easy to drill, and corrosion resistant. They help focus on achieving optimal cement volume, proper plug length, and effective slurry design to enhance long-term wellbore integrity. Their utilization enables the safe placement of long, continuous cement plugs, in some cases exceeding several thousand feet in a single operation, this method has several critical success factors that need to be tailored to be able to produce a high-quality result, especially with this unique challenge of plugging and abandoning two different zones in the same run (Shahril. et al. 2019), without the operational penalties traditionally associated with conventional options. Furthermore, the smooth internal surfaces and low-density nature of fiberglass pipes enhance fluid displacement efficiency during cementing, promoting better cement placement and minimizing the risk of channeling or fluid contamination.

Field deployments have demonstrated that combining plug and abandonment (P&A) plugs with kickoff plugs (KOP) using fiberglass tailpipes not only reduces NPT but also streamlines the transition between abandonment and sidetracking operations. Additionally, fiberglass tailpipes have proven highly effective in curing severe loss zones by enabling the efficient placement of large volumes of cement slurry without risking pipe sticking or placement failure. When designing cement slurry for plug and abandonment operations, long thickening times are essential due to the need to retrieve metallic pipes to the surface. These extended thickening times prevent the slurry from becoming highly gelled, which could restrict pipe movement. Additionally, low gel strength is targeted in the slurry design because, after placing the cement plug, the pipes are pulled out very slowly through the entire length of the plug. This slow movement is considered a low shear moment, creating a static period for the cement plug. During this static period, cement slurry, which is thixotropic in nature, develops gel strength that can hold the pipes in place. To simulate this condition during lab testing, the HPHT consistometer machine is stopped. Using non-metallic, sacrificial pipes simplifies this process significantly, as there is no concern about the pipes getting stuck. This allows for a more fit-for-purpose slurry design without the need for complex formulations.

This approach represents a paradigm shift in how long cement plugs are planned and executed, providing a reliable, efficient, and cost-effective method to ensure well integrity during abandonment and severe loss management. In this paper, the deployment methodology, operational considerations, slurry design optimization, and field performance outcomes associated with this innovative technique will be discussed in detail. The findings will illustrate how the application of fiberglass-based non-metallic tailpipes significantly enhances the efficiency and reliability of cementing operations, particularly in technically challenging environments.

Background and Historical Challenges

Conventional cement plug placement techniques have historically relied on metallic pipes such as drill pipe or work strings to deliver slurry to the intended depth. However, the risk of pipe getting stuck in the setting cement or difficulties in retrieving the pipe led to operational limitations. Typically, this metallic tailpipe had to be retrieved immediately after cement placement with controlled POOH speed and considering the disconnection time, which imposed strict time constraints, leading to pre and controlled gelation and limited plug length to avoid the risk of premature setting issues.

As a result, abandonment of long open hole sections often requires multiple short plugs of 500 ft cement length and more spacer volume is between them, increasing operational complexity, rig time and overall cost. Similarly, curing severe losses during drilling often demanded multiple cementing attempts, leading to high material costs and delayed delivery schedules. The limitations of metallic tailpipes in loss-prone formations also introduce risks of incomplete plug coverage and failure to achieve zonal isolation.

Engineering Approach

In challenging environments, particularly during abandonment or cure loss management across flow potential zones in multiple sections, optimizing the cement plug placement technique is critical to ensure proper cementing operations and achieve job objectives. The introduction of fiberglass non-metallic tailpipes, combined with the appropriate cement slurry design and spacer recipe, offers a transformative solution for these cement plug operations, significantly enhancing placement efficiency, reliability, and operational simplicity.

Optimizing Cementing Performance Using Fiberglass-Based Non-Metallic Tailpipes

The application of fiberglass-based Tailpipe during cement plug placement solves several long-standing technical challenges:

- Risk elimination in case of leaving the fiberglass pipe and avoiding stuck pipe events and associated NPT.
- Extended cement plug lengths, which can confidently place longer continuous plugs, significantly improve zonal isolation and mechanical integrity.
- Drillability. If required in case of a stuck pipe issue, this pipe can be easily drilled through with standard bottomhole assemblies during future sidetracking or re-entry operations without specialized tools and additional runs.
- Operational simplification and smooth internal diameter of fiberglass pipe with preferred size selection as shown in Table 1.0 to enhance fluid displacement during cementing, improving mud removal, and minimizing the risk of contamination or channeling.
- Cost and time optimization by reducing rig time, material waste, repeat operations, and financial benefits compared to conventional approaches.

Table 1.0—Recommended tailpipe size required for a given equivalent open hole size for optimum displacement

Open Hole Size (in)	Preferred Size (in)	Alternative Size (in)
5-⅞	2-⅞	2-⅜
8-⅜	3-½	4
12	5	5-½
>12	5-½	5

The overall impact is a major enhancement in cement plug reliability, reduced operational complexity, and a significant improvement in the economics of plugging and abandonment campaigns or severe losses during drilling.

Cementing Design Considerations and Best Practices

The success of placing long cement plugs using fiberglass-based Non-Metallic Drillable Pipes depends not only on the mechanical advantages of the pipe but also on a highly disciplined cementing design and execution strategy. This section outlines the critical design elements, fluid selection, displacement techniques, and operational practices necessary to ensure plug success, durability, and wellbore integrity.

Cementing Design Considerations and Best Practices.

a. *Thickening Time Control*

- **Pumpability:** The slurry must be designed with a thickening time of the minimum requirement with 2 hours safety margin, allowing for safe placement without the risk of premature setting and stuck pipes, causing an operational complexity.
- In general, the thickening time should be adequate for placement, but strength should develop rapidly. Slurry sedimentation or free water must be scrupulously avoided. If weak formations are present, the slurry density should be the minimum necessary to obtain the desired ultimate strength; otherwise, a dense slurry can be beneficial. To preserve the integrity of the plug, the set cement must not shrink**. (Nelson, E.B. et al. Well Cementing)**
- **Temperature modelling**: Accurately estimating the bottom hole circulating temperature (BHCT) is crucial for designing cement slurry. This involves accounting for dynamic temperature changes during fluid circulation, including frictional heat and cooling effects during placement of the cement slurry to the desired depth. The simulator must consider the thermal conductivity of geological formations, properties of drilling fluids and cement slurry, wellbore geometry, circulation rates, and environmental conditions.

  Additionally, it should model transient heat transfer to reflect evolving wellbore conditions during the cementing job.
- **Temperature variations:** Factors impacting temperature variation in cement plugs should be examined using the simulator. Larger pipes allow more heat transfer due to increased surface area, resulting in faster temperature recovery as compared to smaller pipes as shown in Fig-1 (a) & (b). Fluid viscosity plays a role; thinner fluids promote quicker temperature rise through higher convection, while thicker fluids reduce heat exchange. Pulling speed affects heating; slower pulling allows more time for fluids to heat up, leading to higher final cement temperature, whereas faster pulling results in less heating. Fluid composition, including spacer, mud, and cement, influences temperature variation due to differing heat capacities and viscosities, affecting how quickly fluids warm up or cool down

b. ***Slurry Density Selection***

- For highly fractured, weak formations, lighter slurries (~12.5–13.5 ppg) using lightweight additives may be preferable if well conditions allow.
- For mechanical stability plugs (e.g., KOP or P&A barrier plugs), heavier slurries (16.0–16.5 ppg) provide improved mechanical support and reduce slurry contamination for better zonal isolation.

c. **Slurry Design Criteria**

- Reduce API Fluid Loss (100-300 mL/30 min): Critical to prevent fluid invasion into loss zones or porous rocks, reducing the risk of slurry dehydration and plug porosity, considering very low fluid loss can delay your compressive strength development
- Free Fluid Content (<1%): Essential to avoid slurry separation during long static periods. And zero for horizontal cementing applications.

- Anti-Gas Migration: Targeting the static gel strength (SGS) transition time of less than 30 mins is a critical property to prevent gas migration during the curing period, particularly when placing plugs across the high gas potential zone, especially if any cut in mud weight was observed whilst drilling the OH section.

d. **Slurry Rheology and Stability**

- In practice, rheology must be tailored to well conditions to meet operational needs, as the balanced rheological profile ensures successful application and helps for the cement plug setting and placement, plug integrity, and long-term zonal isolation.

e. **Special Additives (Optional based on Objective)**

- **Expandable and flexible slurry Systems:** Polymer-modified systems that can swell upon exposure to hydrocarbons for P&A long-term integrity.
- **Expandable Cement Slurry System:** Controlled expansion (up to 0.5–2%) to counteract shrinkage and improve bonding.
- **Enhanced Stress Resistance Slurry System**: For wells exposed to CO_2, H_2S, or other corrosive environments.

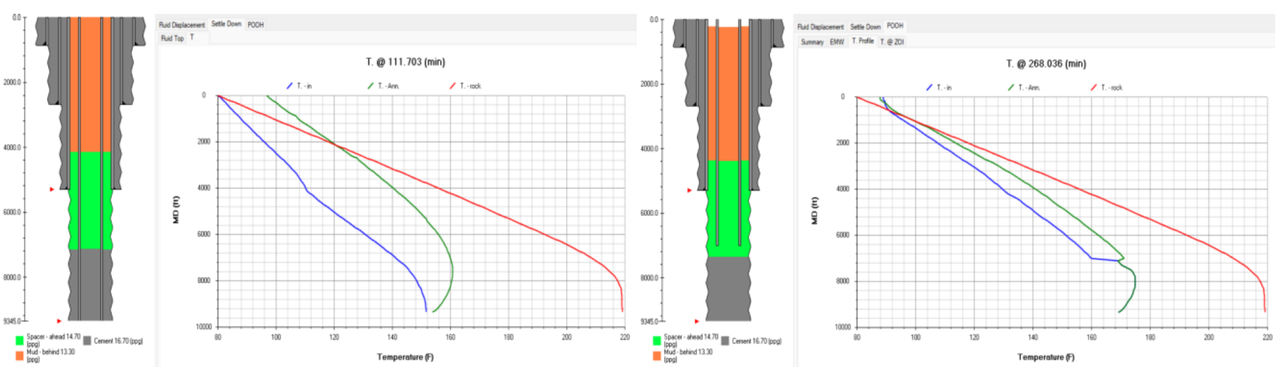

Figure 1 (a)—left side showing 5.5`` drill pipe inside 8.5`` open hole & change in temperature during cement plug placement and right-side with similar pipe configuration showing change in temperature during POOH (blue curve – injection temp, green curve – annular temperature and red curve – geothermal temp)

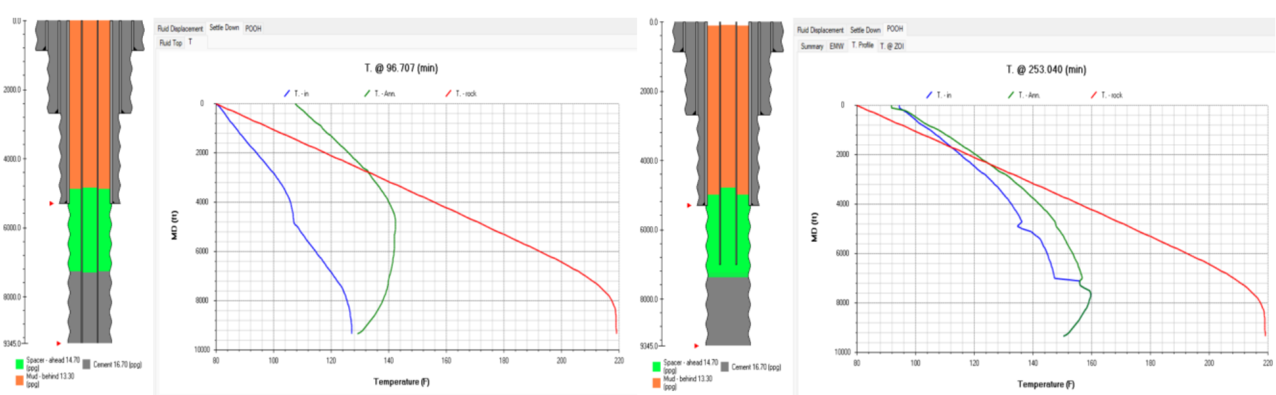

Figure 1 (b)—left side showing 3.5`` drill pipe inside 8.5`` open hole & change in temperature during cement plug placement and right-side with similar pipe configuration showing change in temperature during POOH (blue curve – injection temp, green curve – annular temperature and red curve – geothermal temp)

Spacer and Mud Removal Strategy

Density and Rheology Hierarchy. Establishing a proper density hierarchy is essential to prevent intermixing of fluids during displacement. The spacer fluid should have a density at least 10% higher than the drilling mud and lower than the cement slurry to minimize contamination between fluids.

The Viscosity of the spacer should be higher than that of the drilling mud but lower than the cement slurry. This rheological profile ensures effective mud displacement while maintaining flow stability. Modeling spacer rheology using the Herschel-Bulkley model can aid in predicting flow behavior and optimizing displacement efficiency.

Figure 3 is a sample output from software modeling to confirm the appropriate hierarchy between the fluids at the designed densities and displacement rate essential for effective displacement of bulk drilling fluid from the annulus.

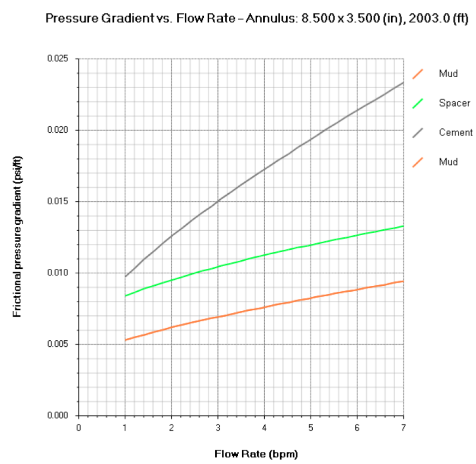

Figure 3—Fluid friction pressure and hierarchy chart.

Spacer Volume and Ensuring Adequate Hole Cleaning. Determining the appropriate spacer volume is a must for effective mud removal. A general guideline is to use a spacer length of +/- 1000 feet or an excess volume of 30% based on the hole volume. This ensures sufficient contact time and flow dynamics to clean the wellbore effectively. Figure 4 is Contact time, which is the period of time that a fluid flows past a particular point in the annular space during displacement. Ideally, the minimum spacer contact time is 10 minutes.

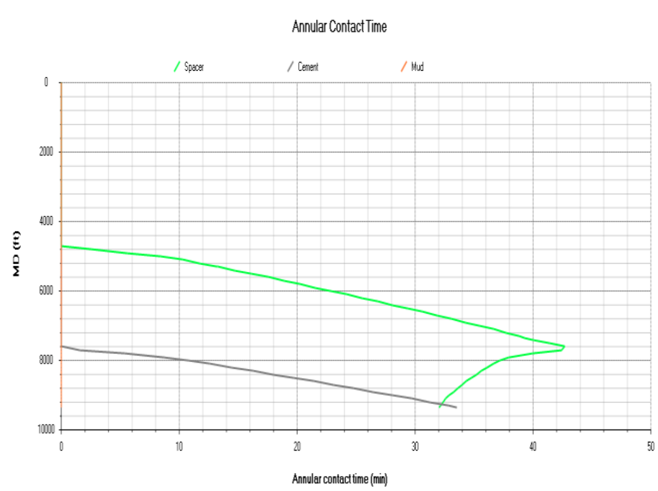

Figure 4—Annular contact time chart.

The spacer system rheological properties can be adjusted depending on the specific scenario, higher density spacers will require less concentration of the spacer mixture, and lower densities will require higher concentration of spacer concentrate. An effective emulsifier system with a tailored surfactant package and a mutual solvent are necessary to enhance removal and ensure effective zonal isolation, This step helps in altering the wettability of the casing and formation surfaces from oil-wet to water-wet, enhancing cement bonding. Inclusion of the added weighting agent will also contribute to the rheological properties.

Fluids compatibility is run in the laboratory to measure the rheological readings of mud-spacer and spacer-cement mixtures. Then the shear stress vs shear rate is plotted to evaluate the fluids compatibility, as can be observed in Table 2 (a) and (b) and Figure 5.

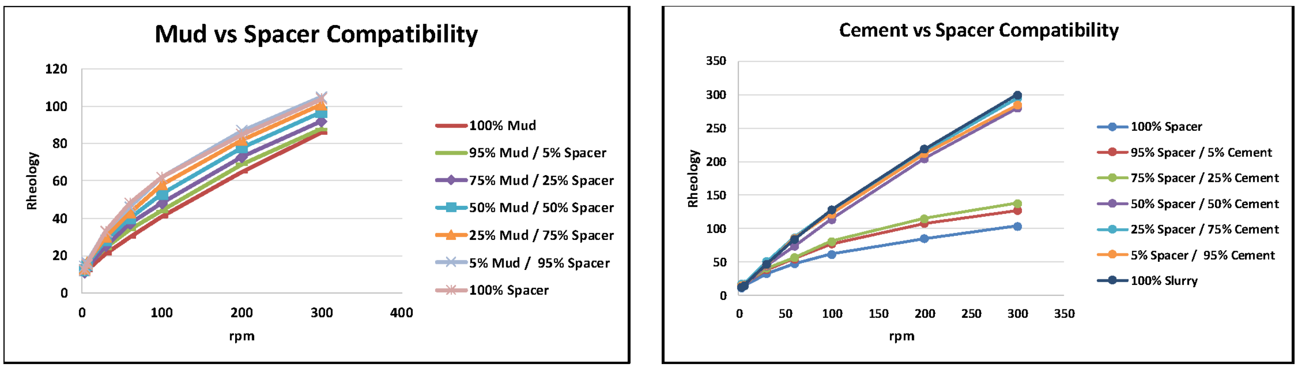

Figure 5—Left: drilling fluid vs spacer compatibility.

Table 2 (a)—Drilling fluid vs spacer compatibility, rheology profiles represented for case study 1.

Mud & Spacer Admixture Rheologies									
Rheology	Mud	Fluid Ratios					Spacer	Compatibility Results	Comments
RPM	100% Mud	95% Mud / 5% Spacer	75% Mud / 25% Spacer	50% Mud / 50% Spacer	25% Mud / 75% Spacer	5% Mud / 95% Spacer	100% Spacer		
300	86	88	92	97	101	105	104	Compatible	No flocculation/gelation, Settling or Separation
200	65	69	73	78	82	87	85	Compatible	
100	41	44	48	53	58	62	62	Compatible	
60	30	34	37	40	43	46	48	Compatible	
30	21	24	25	28	30	33	33	Compatible	
6	12	14	13	15	16	16	15	Compatible	
3	10	11	11	12	13	14	12	Compatible	

Table 2 (b)—Spacer vs cement slurry compatibility, Rheology profiles represented for case study 1.

Spacer & Cement Slurry Admixture Rheologies									
Rheology	Spacer	144					Slurry	Compatibility Results	Comments
RPM	100% Spacer	95% Spacer / 5%	75% Spacer / 25% Cement	50% Spacer / 50% Cement	25% Spacer / 75% Cement	5% Spacer / 95% Cement	100% Slurry		
300	104	127	138	280	295	285	300	Compatible	No flocculation/gelation, Settling or Separation
200	85	108	115	205	215	212	219	Compatible	
100	62	77	81	114	127	122	128.5	Compatible	
60	48	55	57	74	87	86	84	Compatible	
30	33	39	41	45	51	46	47.5	Compatible	
6	15	18	18	15	18	16	15.5	Compatible	
3	12	15	16	14	17	15	13	Compatible	

Right: spacer vs cement slurry compatibility. In both graphics the rheology vs RPM are represented for Field Example 1 below.

Space stability is tested at surface and downhole conditions to ensure adequate rheological properties to sustain solids suspension.

Additional Considerations

a. **Enhanced Spacer System:** Optimizing the right recipe using the appropriate spacer polymer additive can enhance the spacer's performance. These additives improve the spacer's rheological properties and its ability to remove residual mud.
b. **Engineered Lost Circulation Spacer:** Designed to address lost circulation issues and wellbore instability during cementing operations. These spacers form an impermeable barrier over the formation surface, effectively sealing microfractures and minimizing fluid invasion. This barrier not only prevents cement losses but also enhances mud removal by improving displacement efficiency.
c. **Engineered Scrubbing Spacer:** Utilizing spacers with scrubbing capabilities, such as those containing engineered fibers, can significantly improve mud removal efficiency. These spacers physically scrub the wellbore surfaces, aiding in the removal of stubborn mud residues.

Additional techniques

Pump Rate Control and Minimizing ECD Spikes. Fiberglass tailpipes, while advantageous in many respects, have a lower tolerance for pressure fluctuations compared to metallic pipes. To protect the structural integrity of fiberglass components, it's imperative to manage Equivalent Circulating Density (ECD) during cement placement.

Pumping a Low Rheology Mud Ahead. Introducing a low-viscosity mud ahead of the spacer can aid in reducing frictional pressures and ECD. This practice facilitates smoother displacement and minimizes the risk of formation breakdown for long cement plugs.

Mechanical Separation Using Foam Balls. Deploying foam balls ahead of the cement slurry serves as an effective mechanical separator, provided proper sizing is utilized. The selection of the foam wiper ball size should be approximately 10-15% larger than the internal diameter of the work string pipe or tubing to ensure effective wiping without causing excessive resistance.

Field Examples

Field Example 1: Extended Kickoff Plug Placement in Highly Deviated Open Hole

In a highly deviated Oil producer well where the objective was to place a long cement plug of 2000 ft across the 8.5" OH to serve as both a Plug and Abandonment (P&A) barrier and a Kickoff Plug (KOP). The original plan was to place a P&A cement plug, tag it to confirm placement, and then spot a KOP requiring two separate cementing runs. To optimize rig time and eliminate the need for sequential operations, a decision was made to run-in-hole (RIH) 2,003 ft of 3½ in fiberglass tailpipe as a sacrificial, drillable tailpipe, enabling a long cement plug. The cement system used 16.7 ppg with a thickening time exceeding 6.5 hours, and was designed to maintain pumpability throughout the complex interval, which extended from 9,345 ft to 8,015 ft MD. A 140 bbl spacer ahead and a 52 bbl spacer behind were used, each designed at 14.7 ppg to match the rheology and density hierarchy required for efficient 13.3 ppg OBM displacement. A foam wiper ball was introduced ahead of the cement slurry to ensure mechanical separation. Pumping was conducted at a steady 4 bpm, and pipe movement was controlled post-job, during pull-out-of-hole (POOH) at 15 ft/min to minimize surge/swab effects to TOC, then resume POOH to 1,000 ft above cement at normal speed. After reverse circulation to ensure no cement was left inside the pipe, the plug was tagged successfully at the desired TOC depth +/- 8,040 ft MD, confirming the right cement placement and engineered slurry system were used in order to reduce contamination.

Post-job simulation indicated minimal slurry contamination risk after POOH, with most of the cement plug interval falling in the "green" zone, confirming high-quality placement. The tagging depth aligned precisely with the clean cement region, validating the displacement efficiency and the benefit of using the fiberglass tailpipe to place the extended KOP cement in a single operation. Contamination was limited to the upper 773 ft, primarily outside the target isolation interval, thus not affecting the overall plug integrity

The fiberglass pipe enabled extended plug placement in one run and reduced rig time and cost, with zero non-productive time (NPT), and helped a lot to achieve the objective for this application, a clear demonstration of the reliability of fiberglass-based non-metallic pipe in deviated and horizontal wells requiring long plugs for different job objectives.

Field Example 2: High-Density Bottomhole LCP to Isolate High Flow Potential Zone

Drilling a 8⅜" OH section of a vertical gas well exposed the operation to extreme losses and the threat of uncontrolled flow from a highflow potential zone in the field. Due to the severity of the losses and the risk of fluid influx, it was determined that a heavy-weight Loss Circulation Plug (1200 ft LCP) needed to be placed at the bottom of the open hole at 10,510 ft MD/TVD with a tight density operational window to seal the zone and re-establish well control effectively.

To execute this cement plug, 2,000 ft of 2⅜" fiberglass tailpipe (preferred Tailpipe OD size not available in the market) was run to the bottom. A 19.5 ppg heavyweight cement slurry was selected for its ability to resist inflow pressure and provide the mechanical integrity needed to consolidate the fractured formation.

Ahead of the cement, a 100 bbl LCM pill loaded at 150 ppb was pumped to initiate fracture bridging, followed by an engineered LC spacer at 18.70 ppg to help seal microfractures and reduce slurry invasion. Spacer volumes included 80 bbl ahead and 20 bbl behind the cement. A foam wiper ball was used to ensure clean fluid interfaces, followed by 80 bbl of batch-mixed cement placed at a controlled rate of 3 bpm.

Full returns were observed during the displacement, and the string was pulled out at 20 ft/min due to aggressive thickening time that has been considered for this job < 5 hrs with less than one hour safety margin, and to squeeze cement into the fractures. A reverse circulation flush was then carried out to clean the work string of residual cement.

One of the key challenges was the unstable and fluctuating loss rate, which varied between 10%, 50%, and up to 60% during pre-job circulation. This made accurate TOC prediction extremely difficult, as the actual losses could not be anticipated. Multiple TOC simulations were run to reflect each loss scenario, allowing for contingency planning and volume optimization.

Despite the partial losses, the cement plug was successfully tagged and confirmed in place after the WOC period. The use of fiberglass tailpipe was essential in enabling the placement of a long, high-density cement column directly on the bottom without risking a stuck pipe.

The operation successfully isolated the high-flow zone, restoring well stability and allowing the project to move forward without the need for costly sidetracks or additional remediation.

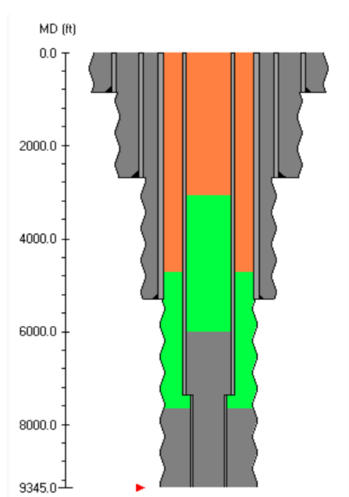

Figure 6—Field Example #1 Well Schematic

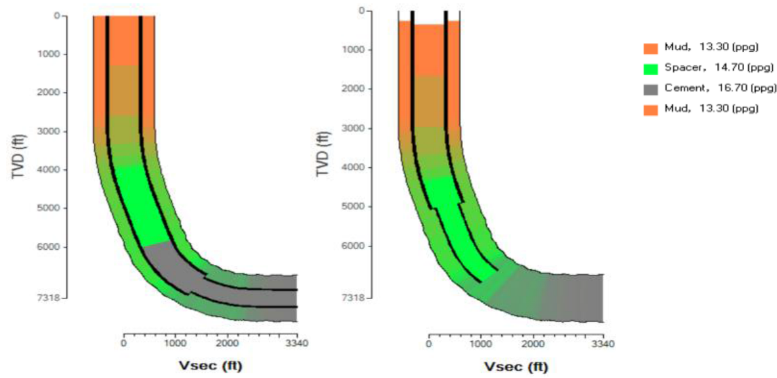

Figure 7—Field Example #1 Contamination risk after POOH (2D)

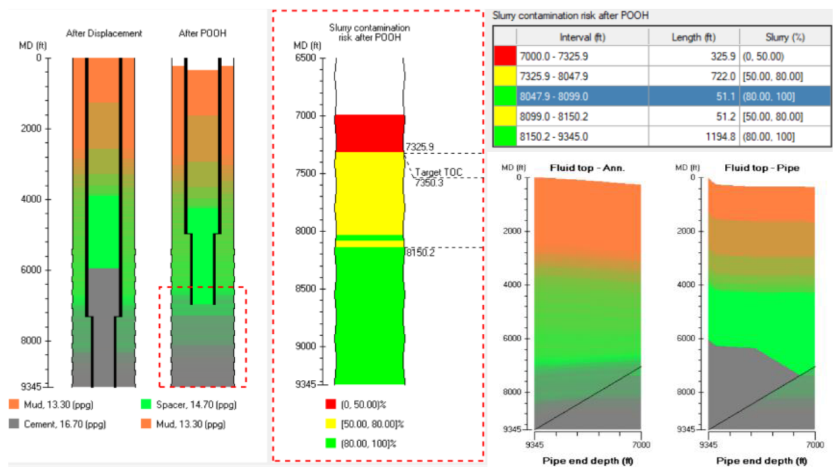

Figure 7—Field Example #1 Contamination risk after POOH (2D) & cement tops before and after in annulus and inside tailpipe

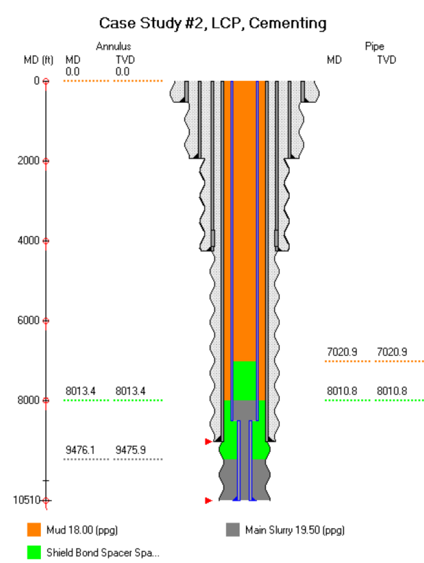

Figure 8—Field Example #2 Well Schematic

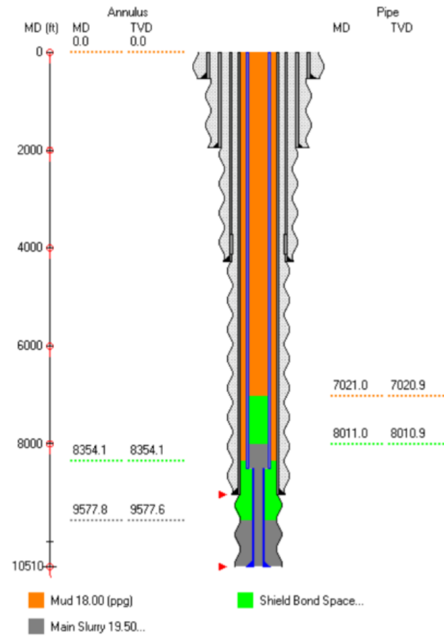

Figure 9—TOC prediction with 10% Losses

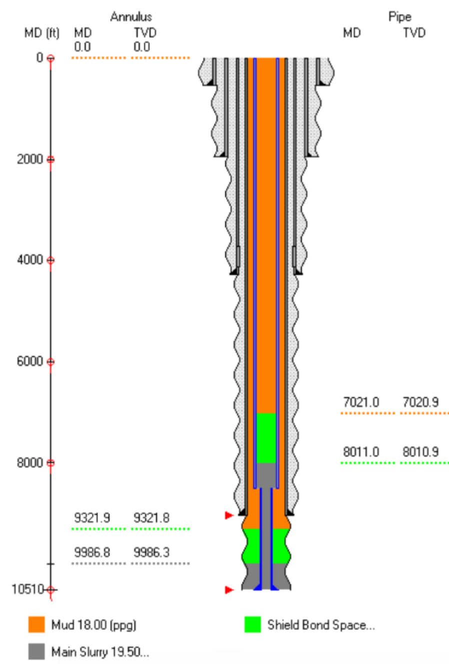

Figure 10—TOC prediction with 50% Losses

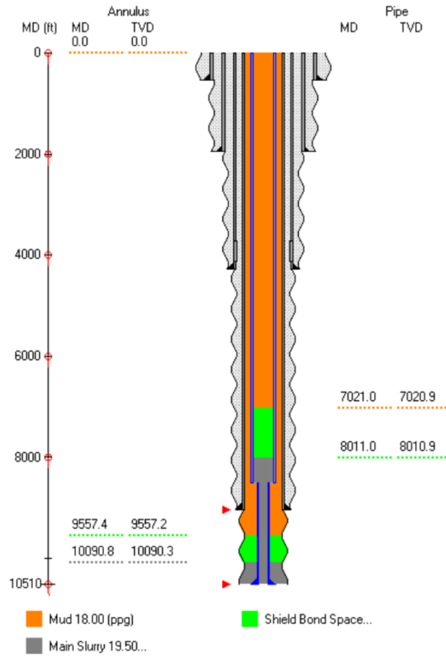

Figure 11—TOC prediction with 60% Losses

Conclusion

The use of fiberglass-based, non-metallic tailpipes offers a game-changing solution for placing long cement plugs in open hole environments, especially during abandonment and loss circulation cement plugs, and reduces stuck pipe risk, enabling longer, more stable cement plugs in a single run.

Field applications have shown significant improvements in operational efficiency, reduced non-productive time (NPT), and enhanced cement placement quality, even in severe loss or high-temperature wells.

Overall, fiberglass tailpipes represent a reliable and cost-effective advancement in modern cementing operations, improving well integrity and simplifying execution in complex wellbores.

Nomenclature - ALL

API	American Petroleum Institute
BHA	Bottom Hole Assembly
P&A	Plug and Abandonment
KOP	Kickoff Plug
OH	Open Hole
NPT	Non-Productive Time
OBM	Oil-Based Mud
WBM	Water-Based Mud
BHST	Bottom Hole Static Temperature
BHCT	Bottom Hole Circulating Temperature
TT	Thickening Time
ECD	Equivalent Circulating Density
PV	Plastic Viscosity
YP	Yield Point
LCP	Loss Circulation Plug
LCM	Lost Circulation Material
NMDP	Non-Metallic Drillable Pipe

ID	Inner Diameter
OD	Outer Diameter
WOC	Wait on Cement
bbl	Barrel (1 bbl = 42 US gallons or ~159 liters)
rpm	Revolutions Per Minute
Bc	Bearden Consistency
BHP	Bottom Hole Pressure
ECD	Equivalent Circulating Density
FG	Fracture Pressure
HPHT	High Pressure High Temperature
OH	Open Hole
RIH	Run in Hole
SG	Specific Gravity
TVD	True Vertical Depth
UCA	Ultrasonic Compressive Strength Analyzer
BHST	Bottom Hole Circulating Temperature
WOC	Waiting on Cement
PPG	Pound per Gallon

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